

UNIT I: DC CIRCUITS, NETWORK THEOREMS & SINGLE-PHASE AC CIRCUITS

Weightage: 25% | Syllabus Section 1

1. Circuit Elements, KCL & KVL

Elements: R (dissipates energy: $P = I^2 R$), L (stores magnetic energy: $E = \frac{1}{2} L I^2$), C (stores electrostatic energy: $E = \frac{1}{2} C V^2$).

Kirchhoff's Laws:

• **KCL (Conservation of Charge):** $\sum I_{\text{in}} = \sum I_{\text{out}}$ at any node.

• **KVL (Conservation of Energy):** $\sum V = 0$ around any closed loop.

Source Transformation: Voltage source (V_s in series with R) $\Leftrightarrow$ Current source ($I_s = V_s/R$ in parallel with R).

2. Core DC Network Theorems

Superposition: Response in linear bilateral circuit with multiple sources equals algebraic sum of individual responses with others deactivated (Voltage $\rightarrow$ Short, Current $\rightarrow$ Open).

Thevenin's Theorem: Any two-terminal linear network $\equiv$ Voltage source V_{th} (Open-circuit voltage) in series with R_{th} .

Norton's Theorem: $\equiv$ Current source I_N (Short-circuit current) in parallel with $R_N = R_{th}$.

Max Power Transfer: Maximum power delivered to load R_L when $R_L = R_{th}$. Max power: $P_{\text{max}} = \frac{V_{th}^2}{4R_{th}}$ (Efficiency = 50%).

3. AC Circuits & Series RLC Resonance



Values: $V_{\text{rms}} = \frac{V_m}{\sqrt{2}} \approx 0.707 V_m$; Form factor = 1.11.

Series RLC Resonance: Occurs when $X_L = X_C$.

- Resonant Freq: $f_0 = \frac{1}{2\pi\sqrt{LC}}$.
- Impedance is minimum: $Z = R$; Current is maximum.
- Quality factor: $Q = \frac{\omega_0 L}{R} = \frac{1}{\omega_0 C R}$.

Parameter / Circuit	Pure Resistive (R)	Pure Inductive (L)	Pure Capacitive (C)
Phase Relation (ϕ)	Voltage & Current are strictly in phase ($\phi = 0^\circ$).	Current lags voltage by 90° ($\phi = -90^\circ$).	Current leads voltage by 90° ($\phi = +90^\circ$).
Impedance / Reactance	$Z = R$	$X_L = \omega L = 2\pi f L$	$X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C}$
Power Factor ($\cos \phi$)	$\cos(0^\circ) = 1.0$ (Unity)	$\cos(90^\circ) = 0.0$ (Lagging)	$\cos(90^\circ) = 0.0$ (Leading)

UNIT II: TRANSFORMERS & ELECTRICAL ROTATING MACHINES

Weightage: 25% | Syllabus Section 2

1. Single-Phase Transformer: Working & Testing

Principle: Mutual electromagnetic induction. EMF equation: $E_1 = 4.44 f N_1 \Phi_m$; $E_2 = 4.44 f N_2 \Phi_m$.

Transformation Ratio: $K = \frac{V_2}{V_1} = \frac{N_2}{N_1} = \frac{I_1}{I_2}$.

Losses:

• **Core / Iron Loss (P_i):** Constant; Hysteresis $P_h = k_h f B_m^{1.6}$ + Eddy current $P_e = k_e f^2 B_m^2 t^2$.

• **Copper Loss (P_{cu}):** Variable; $P_{cu} = I_1^2 R_1 + I_2^2 R_2$.

Condition for Maximum Efficiency: $P_{cu} = P_i$ (Variable Copper loss = Constant Iron loss).

Testing Diagnostics:

• **Open Circuit (OC) Test:** Conducted on LV side at rated voltage with HV open. Measures Core Losses (P_i , R_0 , X_0).

• **Short Circuit (SC) Test:** Conducted on HV side with LV shorted at reduced voltage. Measures Copper Loss (P_{cu} , R_{eq} , X_{eq}).

2. DC Machines & Induction Motors

DC Generator EMF: $E_g = \frac{\Phi Z N P}{60 A}$, where A = 2 (Wave) or P (Lap).

DC Motor Torque: $T_a = \frac{1}{2\pi} \frac{\Phi Z I_a P}{A} \approx 0.159 \frac{\Phi Z I_a P}{A}$ N-m.

Speed Equation: $N \propto \frac{E_g}{\Phi} = \frac{V - I_a R_a}{\Phi}$.

• **Speed Control of DC Shunt:** (1) Armature Resistance Control (below rated speed), (2) Field Flux Control (above rated speed).

3-Phase Induction Motor:

• Synchronous Speed: $N_s = \frac{120 f}{P}$.

• Slip: $s = \frac{N_s - N}{N_s}$. Rotor freq: $f_r = s \cdot f$.

• Torque-Slip Curve: At low slip, $T \propto s$ (linear); at high slip, $T \propto 1/s$. Max torque is independent of rotor resistance!

1-Phase Induction Motor: Not self-starting. Explained by *Double Revolving Field Theory* (pulsating field resolves into forward & backward rotating fields). Starting methods: Split-phase, Capacitor-start.

UNIT III: SEMICONDUCTOR DEVICES, RECTIFIERS & BIPOLAR JUNCTION TRANSISTORS

Weightage: 25% | Syllabus Section 3

1. P-N Diode, Rectifiers & Zener Regulator

P-N Junction: Built-in barrier potential: 0.7V (Si), 0.3V (Ge). Diode equation: $I = I_0 \left(e^{\frac{V}{V_T}} - 1 \right)$.

Parameter	Half-Wave (HWR)	Full-Wave Center-Tap	Bridge Rectifier
No. of Diodes	1	2	4
Max Efficiency (η)	40.6%	81.2%	81.2%
Ripple Factor (γ)	1.21	0.482	0.482
Peak Inverse (PIV)	V_m	$2V_m$	V_m

Zener Diode: Heavily doped; operates in reverse breakdown (Zener/Avalanche). Used as Shunt Voltage Regulator maintaining constant V_Z across load.

Clippers & Clampers: Clipper limits/cuts off voltage levels; Clamper shifts entire AC signal to a different DC level without changing waveform shape.

2. BJT Configurations & Operating Regions

BJT Construction: Emitter (heavily doped, medium size), Base (lightly doped, ultra-thin), Collector (moderately doped, largest size to dissipate heat).

Current Relation: $I_E = I_B + I_C$. Current Gains: $\alpha = \frac{I_C}{I_E}$ (0.95–0.99), $\beta = \frac{I_C}{I_B}$ (20–500).
Relation: $\beta = \frac{\alpha}{1-\alpha}$.

Operating Region	Emitter-Base Junc	Collector-Base Junc	Application
Cutoff	Reverse Biased	Reverse Biased	Open Switch (OFF)
Active	Forward Biased	Reverse Biased	Linear Amplifier
Saturation	Forward Biased	Forward Biased	Closed Switch (ON)

CE Characteristics: Most popular configuration providing both high voltage and current gain (High Power Gain).

UNIT IV: DIGITAL ELECTRONICS, BOOLEAN ALGEBRA & K-MAP MINIMIZATION

Weightage: 25% | Syllabus Section 4

1. Number Systems & Codes

Bases: Binary (2), Octal (8), Decimal (10), Hexadecimal (16: 0-9, A-F).

Binary Arithmetic:

- 1's Complement: Invert bits ($0 \rightarrow 1, 1 \rightarrow 0$).
- 2's Complement: 1's Complement + 1.
- Subtraction: $A - B = A + (2\text{'s comp of } B)$.

Codes:

- BCD (8421): Each decimal digit encoded into 4 bits.
- Excess-3: BCD + 0011 (Self-complementing).
- Gray Code: Unit-distance code (only 1 bit flips between consecutive values: $G_i = B_i \oplus B_{i+1}$).

2. Logic Gates & De Morgan's Laws

Universal Gates: NAND & NOR (can implement AND, OR, NOT, XOR, XNOR).

Exclusive Gates:

- XOR: $Y = A \oplus B = A\bar{B} + \bar{A}B$ (Odd parity).
- XNOR: $Y = A \odot B = AB + \bar{A}\bar{B}$ (Equivalence).

De Morgan's Theorems:

- $\overline{A + B} = \bar{A} \cdot \bar{B}$ (Break line, change sign)
- $\overline{A \cdot B} = \bar{A} + \bar{B}$

Canonical Forms: SOP (Sum of Minterms Σm) and POS (Product of Maxterms ΠM).

3. 4-Variable Karnaugh Map (K-Map)

Rules: Group adjacent 1s in powers of 2 (1, 2, 4, 8, 16). Larger groups eliminate more variables.

	00	01	11	10
00				
01				
11				
10				

Don't Care Conditions $\frac{d}{x}$: Used as 1s if they help form larger groupings; otherwise ignored as 0s to minimize final gate count.

Decimal	Binary (4-Bit)	Octal	Hexadecimal	BCD (8421)
5	0101	5	5	0000 0101
10	1010	12	A	0001 0000
15	1111	17	F	0001 0101